

Experiment 'p_4_1_b_sum_squares' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_1_b_sum_squares

Natural language statement: Write a method with loops that computes the sum of all squares between 1 and 100 (inclusive).

Method signature: p_4_1_b_sum_squares() returns (sum: int)

Ensures

- `sum == 338350`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method with loops that computes the sum of all squares between 1 and 100 (↪ inclusive).

The signature should be:

`p_4_1_b_sum_squares()` returns (sum: int)

The method should respect the following contract:

ensures sum == 338350

Produce and show only the Dafny body of this method, including the curly braces that ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdxUVVszZRP2338lQ8GpnpFhRf3j

Model: gpt-5.5-2026-04-23

Created at: 1780485716

Finish reason: stop

Usage: 266 tokens in, and 6552 tokens out

```
{
  sum := 0;
  var i := 1;
  while i <= 100
    invariant 1 <= i <= 101
    invariant i == 1 ==> sum == 0
    invariant i == 2 ==> sum == 1
    invariant i == 3 ==> sum == 5
    invariant i == 4 ==> sum == 14
    invariant i == 5 ==> sum == 30
    invariant i == 6 ==> sum == 55
    invariant i == 7 ==> sum == 91
    invariant i == 8 ==> sum == 140
    invariant i == 9 ==> sum == 204
    invariant i == 10 ==> sum == 285
    invariant i == 11 ==> sum == 385
    invariant i == 12 ==> sum == 506
    invariant i == 13 ==> sum == 650
    invariant i == 14 ==> sum == 819
    invariant i == 15 ==> sum == 1015
```

```
invariant i == 16 ==> sum == 1240
invariant i == 17 ==> sum == 1496
invariant i == 18 ==> sum == 1785
invariant i == 19 ==> sum == 2109
invariant i == 20 ==> sum == 2470
invariant i == 21 ==> sum == 2870
invariant i == 22 ==> sum == 3311
invariant i == 23 ==> sum == 3795
invariant i == 24 ==> sum == 4324
invariant i == 25 ==> sum == 4900
invariant i == 26 ==> sum == 5525
invariant i == 27 ==> sum == 6201
invariant i == 28 ==> sum == 6930
invariant i == 29 ==> sum == 7714
invariant i == 30 ==> sum == 8555
invariant i == 31 ==> sum == 9455
invariant i == 32 ==> sum == 10416
invariant i == 33 ==> sum == 11440
invariant i == 34 ==> sum == 12529
invariant i == 35 ==> sum == 13685
invariant i == 36 ==> sum == 14910
invariant i == 37 ==> sum == 16206
invariant i == 38 ==> sum == 17575
invariant i == 39 ==> sum == 19019
invariant i == 40 ==> sum == 20540
invariant i == 41 ==> sum == 22140
invariant i == 42 ==> sum == 23821
invariant i == 43 ==> sum == 25585
invariant i == 44 ==> sum == 27434
invariant i == 45 ==> sum == 29370
invariant i == 46 ==> sum == 31395
invariant i == 47 ==> sum == 33511
invariant i == 48 ==> sum == 35720
invariant i == 49 ==> sum == 38024
invariant i == 50 ==> sum == 40425
invariant i == 51 ==> sum == 42925
invariant i == 52 ==> sum == 45526
invariant i == 53 ==> sum == 48230
invariant i == 54 ==> sum == 51039
invariant i == 55 ==> sum == 53955
invariant i == 56 ==> sum == 56980
invariant i == 57 ==> sum == 60116
invariant i == 58 ==> sum == 63365
invariant i == 59 ==> sum == 66729
invariant i == 60 ==> sum == 70210
invariant i == 61 ==> sum == 73810
invariant i == 62 ==> sum == 77531
invariant i == 63 ==> sum == 81375
invariant i == 64 ==> sum == 85344
invariant i == 65 ==> sum == 89440
invariant i == 66 ==> sum == 93665
```

```

invariant i == 67 ==> sum == 98021
invariant i == 68 ==> sum == 102510
invariant i == 69 ==> sum == 107134
invariant i == 70 ==> sum == 111895
invariant i == 71 ==> sum == 116795
invariant i == 72 ==> sum == 121836
invariant i == 73 ==> sum == 127020
invariant i == 74 ==> sum == 132349
invariant i == 75 ==> sum == 137825
invariant i == 76 ==> sum == 143450
invariant i == 77 ==> sum == 149226
invariant i == 78 ==> sum == 155155
invariant i == 79 ==> sum == 161239
invariant i == 80 ==> sum == 167480
invariant i == 81 ==> sum == 173880
invariant i == 82 ==> sum == 180441
invariant i == 83 ==> sum == 187165
invariant i == 84 ==> sum == 194054
invariant i == 85 ==> sum == 201110
invariant i == 86 ==> sum == 208335
invariant i == 87 ==> sum == 215731
invariant i == 88 ==> sum == 223300
invariant i == 89 ==> sum == 231044
invariant i == 90 ==> sum == 238965
invariant i == 91 ==> sum == 247065
invariant i == 92 ==> sum == 255346
invariant i == 93 ==> sum == 263810
invariant i == 94 ==> sum == 272459
invariant i == 95 ==> sum == 281295
invariant i == 96 ==> sum == 290320
invariant i == 97 ==> sum == 299536
invariant i == 98 ==> sum == 308945
invariant i == 99 ==> sum == 318549
invariant i == 100 ==> sum == 328350
invariant i == 101 ==> sum == 338350
decreases 101 - i
{
  sum := sum + i * i;
  i := i + 1;
}
}

```

Final Program

```

method p_4_1_b_sum_squares() returns (sum: int)
  ensures sum == 338350
{
  sum := 0;
  var i := 1;
  while i <= 100

```

```

invariant 1 <= i <= 101
invariant i == 1 ==> sum == 0
invariant i == 2 ==> sum == 1
invariant i == 3 ==> sum == 5
invariant i == 4 ==> sum == 14
invariant i == 5 ==> sum == 30
invariant i == 6 ==> sum == 55
invariant i == 7 ==> sum == 91
invariant i == 8 ==> sum == 140
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invariant i == 36 ==> sum == 14910
invariant i == 37 ==> sum == 16206
invariant i == 38 ==> sum == 17575
invariant i == 39 ==> sum == 19019
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invariant i == 47 ==> sum == 33511
invariant i == 48 ==> sum == 35720
invariant i == 49 ==> sum == 38024
invariant i == 50 ==> sum == 40425

```

```
invariant i == 51 ==> sum == 42925
invariant i == 52 ==> sum == 45526
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invariant i == 98 ==> sum == 308945
invariant i == 99 ==> sum == 318549
invariant i == 100 ==> sum == 328350
invariant i == 101 ==> sum == 338350
```

```
    decreases 101 - i
  {
    sum := sum + i * i;
    i := i + 1;
  }
}
```

Total Token Usage

Input tokens: 266

Output tokens: 6552

Reasoning tokens: 5116

Sum of 'total tokens': 6818

Experiment Timings

Overall Experiment started at 1780485716674, ended at 1780485834685, lasting 118011ms (118.01 seconds)

Iteration #1 started at 1780485716675, ended at 1780485834684, lasting 118009ms (118.01 seconds)